

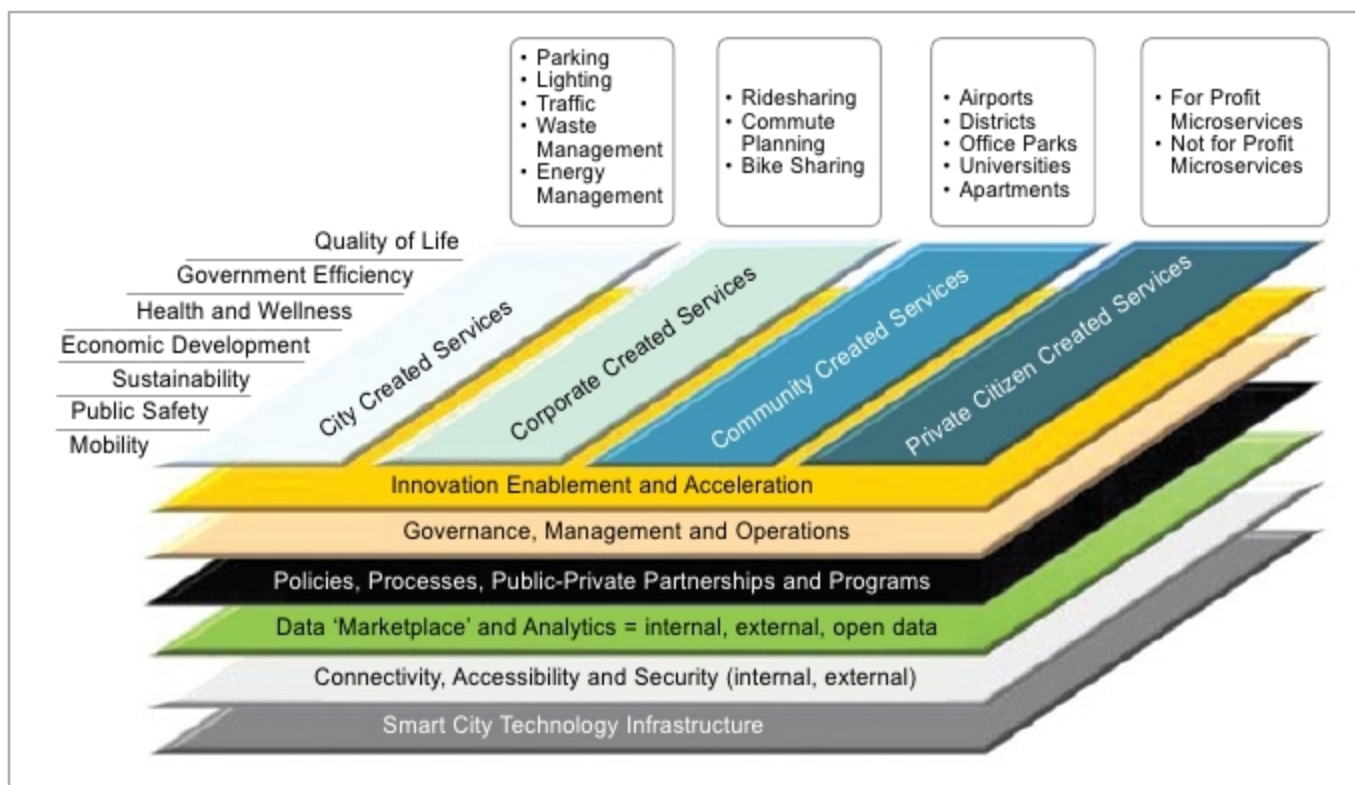


Fig. 3: The smart city ecosystem model (Credit: www.strategyofthings.io)

Initiatives for making smart cities

A prime example here is managing energy use in buildings—heating, air-conditioning and lighting—while maintaining comfortable room temperatures, monitoring facility status like occupancy and availability. These end-to-end solutions offer user-friendly apps on officials' smartphones and provide programmed, automatic rule-based responses to changing environmental or weather conditions. Monitoring and counting the movements and whereabouts of building occupants or citizens in public areas also bring an effective smart-city application. The same way, LoRaWAN's native geolocation capability also enables cities to track their assets.

Sustainability is a critical aspect of smart-city networks. Outdoors, cities can leverage low-power wide-area (LPWA) sensors and networks for traffic monitoring, streetlighting, parking availability, waste collection in addition to noise detection, like gunshots, and environment sensing.

In Kakinada, India, Kerlink's Wirnet TM stations are powering a comprehensive, ambient air-quality monitoring system designed by Oizom that monitors air and noise pollutions and other environmental hazards.

Chengyang is a smart city project established by Huawei. It converts a

wasteland into arable land capable of growing seawater rice using a combination of sensors, wireless and deep learning technologies. It feeds 80 million people using previously unusable land. This innovation is now being applied in other parts of the world.

India Maven, a smart-city solution provider from Pune, provides smart solutions in the fields of water and energy metering, lighting, agriculture, ambulance tracking, medical wearables, firefighting, city-pollution monitoring, smart parking, renewable energy management system and intelligent traffic management.

Dejean adds, "With increasing vital IoT deployments for cities, utilities, airports, harbours, farms, manufacturing plants and other sectors, we have LoRaWAN IoT connectivity building block. We also have industrial-grade network infrastructure—indoor and outdoor carrier-grade range of LoRaWAN gateways—with a full suite of user-centric value-added services that empower cities of any size to efficiently manage their IoT connectivity. The Wanesy Management Center, Kerlink's Wanesy Geolocation and Wanesy Device Management solution value-added services can help cities not only manage/locate their assets and remote sensors, but also help them improve the services they grant to their citizens, like traffic information, parking availability, weather forecast, public-

building attendance and public-private partnerships (PPP)."

Smart City Expo World Congress is looking to establish frameworks that will shape the collaborative approach to establish smart cities of the future. Its theme for 2018 was to determine how to structure smart cities in a future-proof and innovative manner.

International Electro-technical Commission (IEC) Market Strategy Board brings together chief technology officers of leading companies to guide city planners and regulators, and develop standards in an effort to

move cities to greater smartness.

Many places have a number of projects implemented on the ground, but these do not scale up because there are no steps taken at the national level or by individual government bodies. As more and more people are migrating towards cities, it is essential to develop a sustainable environment to make these places worth living. Human behaviour as a city dweller is changing, so everything has to be taken into account, including building material, climate, technology and so on while designing and building the cities of tomorrow.

Conclusion

Smart cities must fulfil the need for sufficient fresh water, universal access to cleaner energy, ability to travel efficiently from one point to another, sense of safety and security, among others. Smart cities need to efficiently combine a smart physical layer (ability to use sensors to connect physical assets like buildings, public transportation, energy grid, etc) with a smart digital layer (increasingly mixing capabilities like Big Data, AI, the IoT and the like) to crunch data collected and turn it into actionable decisions on the field, in a cost-efficient way. These cities need to meet the needs of current and future generations, and to maximise benefits while minimising the negative aspects of their daily living. **EFY**